EXERCISE 10A - Slow Flight

AIM To recognise and recover from flight at critically low airspeeds in balance.

AIRMANSHIP LOOKOUT - HASELL/HELL CHECKS - V_{FE} - P.O.H. - STALL WARNER - VS_I + VS_0

Exercise: Revision. Climb to Altitude, Straight & Level

HASELL

(1) **VS1 + 10kts = 50kts**

Power 1700rpm Carb Heat ON

Attitude – Maintain 50kts Straight & Level

High Angle of Attack / High Drag

More Power may be needed to maintain Straight & Level

Trim – 50kts

Elevator = Speed Power = Altitude

(Change one → Change the other)

(2) **Turns Straight & Level 15˚ Angle of Bank**

Bank Angle Degrades Performance

More Power to Maintain Altitude

Left then **Right**

(3) **Climbing Turn**

Full Power – Maintain 50kts – TRIM

Increased Angle of Bank reduces Rate of Climb

So Trade Off (Back pressure for Angle of Bank)

No More Power to Increase any Speed Loss

Note Rate of Climb

Height – 3000' AGL

Airframe – Clean

Security – Loose Articles

Engine – T's and P's, Carb Heat

Location – ABCD

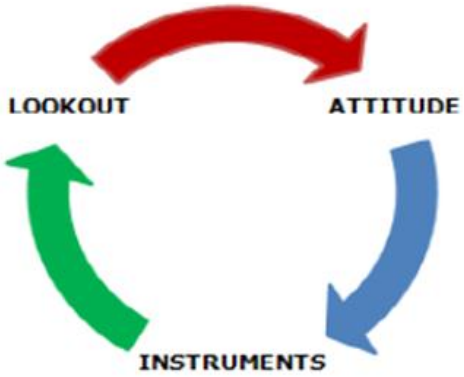
Look Out – 2 x 90˚

Return to Straight & Level

Power 2300rpm CARB HEAT OFF

Attitude = Straight & Level

Trim = 90kts



HELL

(1) **Vs0 + 10kts = 45kts**

Power 1700rpm Carb Heat On

White Arc - 30˚ Flaps (In Stages)

Attitude – Maintain 45kts S+L

Different Picture with Flaps

Increase Power – More Drag

Trim 45kts

(2) **Turns S+L 15˚ A.O.B**

As Before

But Greater Drag/ More Power

Left then **Right**

(3) **Climbing Turns**

Full Power – Maintain 45kts –Trim

Increased A.O.B. Reduces R.O.C

Trade Off. Note R.O.C